User Manual

Smart Vehicle Tracking and Management System



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Smart Vehicle Tracking and Management System

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1. Introduction

- ## 1.1 Welcome to Smart Vehicle Tracking System



Welcome to the Smart Vehicle Tracking and Management System, an innovative IoT-based solution designed to revolutionize vehicle Cargo management and cargo monitoring. This comprehensive system integrates advanced hardware sensors with intuitive software applications to provide real-time tracking, cargo monitoring, alert management, report generation, and operational efficiency for small and medium-sized transportation businesses.

Our system addresses critical challenges faced by cargo operators, including vehicle security, driver safety, cargo monitoring, and operational transparency. By combining GPS tracking, environmental sensors, and emergency detection capabilities, we deliver a complete solution that enhances safety and productivity in vehicle operations.

1.2 System Overview

The Smart Vehicle Tracking and Management System consists of three main components working seamlessly together

- **IoT Vehicle Device:** A robust hardware unit installed in each vehicle, equipped with a GPS module, accelerometer for accident detection, temperature and humidity sensors for cargo monitoring, a GSM communication module, and tamper-detection sensors. This device continuously collects and transmits real-time data about vehicle location, environmental conditions, and safety status.
- **Administrator Web Application:** A comprehensive web-based dashboard that provides cargo managers with complete oversight of vehicle operations. The application enables vehicle and driver management, real-time tracking visualization, task assignment, alert configuration, performance analytics, and detailed reporting capabilities.
- **Driver Mobile Application:** A user-friendly mobile app designed for drivers to interact with the system on the go. The app provides task management, emergency alerts, trip logging, and direct communication with cargo administrators.

All components are interconnected through secure cloud infrastructure, ensuring real-time data synchronization and reliable communication between all system elements.

1.3 User Types and Access Levels

- **Cargo System Administrator:** Has full access to all system features, including company setup, user management, system configuration, advanced analytics, and maintenance functions. Administrators manage the entire Cargo, configure global settings, assign vehicles and drivers, create and monitor tasks, manage alerts, generate comprehensive operational reports, coordinate emergency responses, and access real-time tracking and performance analytics.
- **Driver:** Has limited access focused on operational tasks, including trip management, task updates, and profile management. Drivers interact primarily through the mobile application, which provides role-specific features such as task lists, trip logging, status updates, and emergency alert functions.

1.4 Getting Started Checklist

Before beginning system implementation, ensure the following prerequisites are met:

Hardware Preparation: Verify that all IoT device components are available and functional, including ESP32 microcontroller, GPS module, GSM module, sensors, and power supply components. Ensure proper installation tools and mounting hardware are ready.

Network Requirements: Confirm stable internet connectivity for web application access, GSM network coverage for device communication, and WiFi availability for device configuration and updates.

Account Setup: Prepare company information for administrator registration, including business details, contact information, and initial user accounts. Plan user roles and access levels for different team members

2. System Requirements

• 2.1 Administrator Web Application Requirements

Component	Requirement
Hardware	Desktop/laptop with ≥ 4GB RAM, dual-core processor; monitor ≥ 1366×768 resolution
Operating System	Windows 10+, macOS 10.14+, Linux Ubuntu 18.04+
Web Browser	Chrome (recommended), Firefox, Edge, Safari (latest versions); JS and cookies enabled
Internet	Stable broadband with ≥ 10 Mbps download speed
Additional Software	PDF Reader; Excel or spreadsheet software for report viewing & analysis

• 2.2 Driver Mobile App Requirements

Component	Requirement
Mobile Device	Android 7.0+ or iOS 12.0+; ≥ 2GB RAM and 1GB free storage
Display	Minimum 5-inch touchscreen
Connectivity	4G LTE or Wi-Fi; GPS must be enabled
Permissions	Location services and push notifications enabled
Battery	Sufficient for full shift; portable/vehicle charger recommended

• 2.3 IoT Device Requirements

Component	Requirement
Power Supply	5V DC vehicle power; 3.7V Li-ion backup battery; 2–3W power usage
Environment	Sheltered from weather; −20°C to +70°C; ≤ 95% humidity
Connectivity	GSM with 2G/3G/4G; Wi-Fi for setup/updates; data plan required
Physical Setup	Secure, hidden mount (dashboard/engine bay/cargo); tamper detection recommended
Maintenance Access	Visible LED status; accessible for updates/troubleshooting

3. IoT Device (Vehicle Unit) Guide

The Vehicle Tracking and Management System is shipped as a pre-assembled, plug and play IoT device that can be easily mounted inside any vehicle. All essential sensors, communication modules, and power circuitry are integrated onto a custom PCB, enclosed within a durable 3D printed case. This section provides a complete guide to the hardware, installation, and operation of the in vehicle unit.

3.1 Device Overview

The IoT unit continuously monitors vehicle status, captures environmental and positional data, and transmits it in real time to the backend system via GSM/GPS modules. It is designed to be compact, rugged, and suitable for both commercial and personal vehicles.

3.1.1 Hardware Components

Component	Description
ESP32 Controller	Main microcontroller with Wi-Fi, Bluetooth, and serial communication interfaces.
GPS Module	Provides real time location tracking using satellite signals.
SIM800L GSM Module	Enables data communication and SMS alerts over the cellular network.
ADXL345 Accelerometer	Detects vehicle motion, sudden impacts (accidents), and tampering attempts.
DHT11 Sensor	Measures in-vehicle temperature and humidity.
Buzzer Alarm	Audible alert for tamper or temperature alerts.
LCD Display	Display informations

Power Supply Circuit	5V input powers the TP4056 module, which charges a 3.7V Li-ion battery. A boost converter steps up the 3.7V battery output to 5V, ensuring stable operation of the ESP32 and other modules during power interruptions.
PCB and Connectors	Custom PCB with all modules pre-mounted and plug-in terminals for wiring.

3.1.2 Status Indicators

Indicator	Description
Power LED, GPS Status, Network Status, Temperature Display	ESP32 onboard LED shows power, GPS fix, GSM network, and real-time temperature displayed on LCD.
GSM LED	Shows GSM network connectivity status (blinking = searching, solid = connected).
GPS LED	Indicates GPS satellite lock status.

3.2 Installation Process

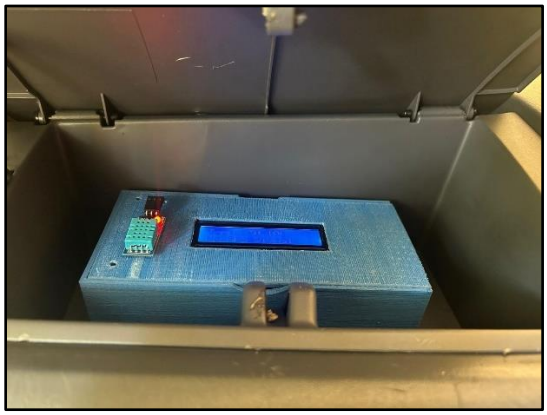
3.2.1 Pre-Installation Checklist

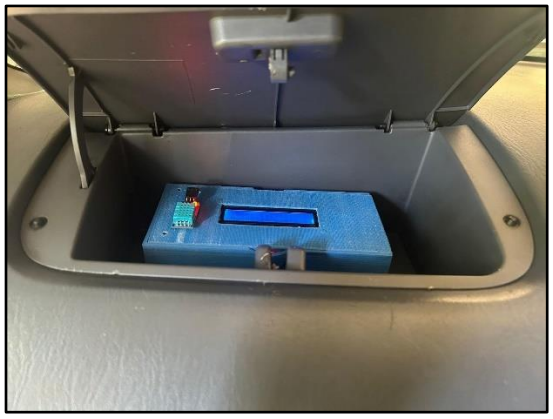
1. Make sure your vehicle provides a stable 5V power supply to power the device.
2. Check that your GSM SIM card is active and has an SMS or data plan.
3. Ensure the GPS antenna will have a clear view, avoid placing it under metal surfaces.
4. Gather all necessary tools and materials, such as wires, zip ties, and mounting clips.
5. Confirm the lithium battery connected to the charger is properly installed and in good condition.

3.2.2 Vehicle Installation

- Mount the device securely inside the vehicle. Suitable locations include under the dashboard, beneath a seat, or any other safe and concealed spot chosen by the user.
- The 3D-printed box has a special compartment to hold the GPS antenna securely and help maintain good signal reception.
- Use zip ties to neatly organize and secure all cables.
- Fix the device enclosure firmly using the mounting holes or brackets provided to prevent movement while driving.

Eg: You can install the device in the vehicle using the method shown below.

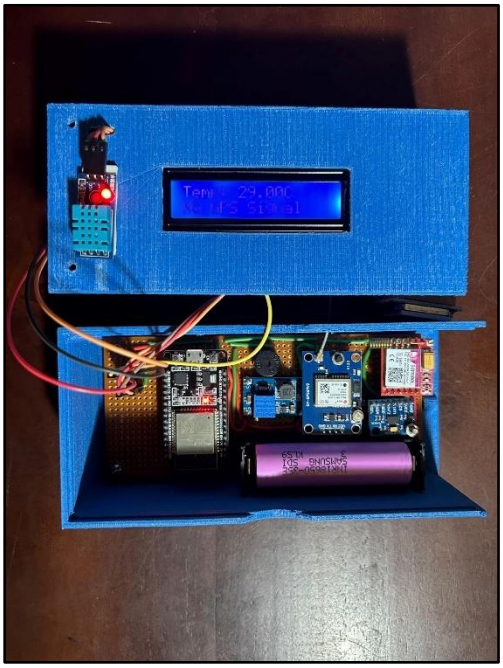




3.2.3 Power Connection

- Connect the vehicle's 5V power supply directly to the TP4056 charger module input.
- If the module works without external power, it can run on the battery for about 4 to 5 hours.

Eg: Device status, with vs. without external power.



without external power



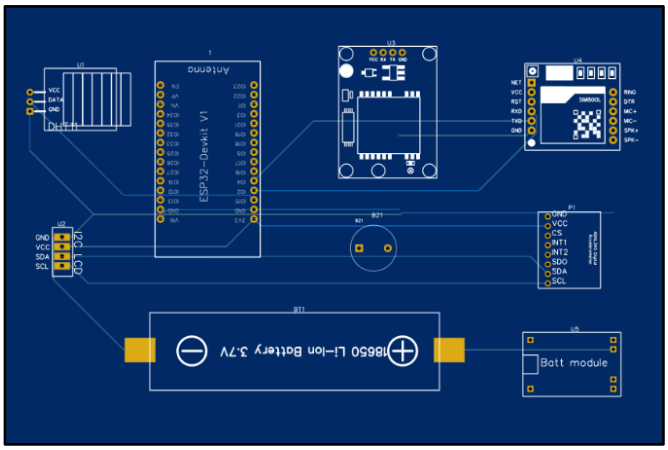
with external power

3.3 Device Configuration

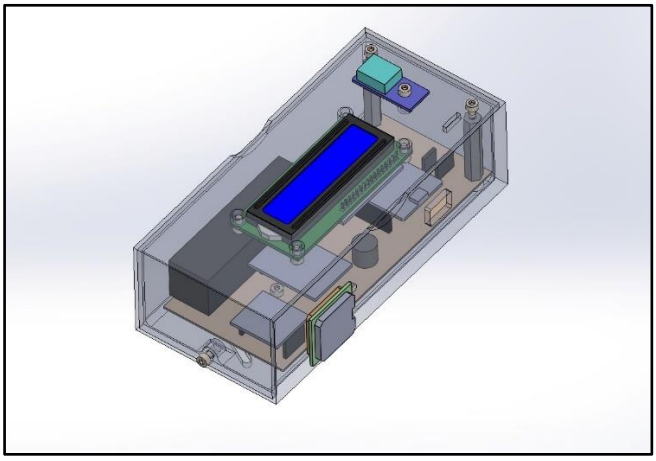
3.3.1 Initial Setup

- Connect the ESP32 to a PC using a USB cable for firmware flashing, if needed.
- Connect the ESP32 to your PC via USB to upload the firmware using an IDE like Arduino IDE or PlatformIO, then configure settings such as Firebase credentials and sensor thresholds through the serial monitor or a setup script.

- Verify that all sensors connected via the custom PCB are working correctly by checking the serial output logs.
  Eg: This PCB design can be used to connect all sensor modules.



Eg:You can also mount this PCB and other components inside the box as shown.



3.3.2 Network Configuration

- Insert the activated SIM card into the SIM800L module mounted on your PCB.
- Configure the GSM settings (APN, server IP, and port) using AT commands, or use the default pre-configured settings.
- Ensure that both GPS and GSM modules are able to receive signals and successfully send data.

- ## 3.4 Device Operation

  - ### 3.4.1 Normal Operation Mode
    - When powered on, the device automatically starts in Normal Operation Mode.
    - It continuously collects data from all connected sensors via the custom PCB.
    - GPS data is updated every few seconds, while temperature and motion sensor readings are taken at preset intervals.
    - The collected data is transmitted periodically to the cloud for monitoring.

  - ### 3.4.2 Data Transmission
    - All sensor data is securely sent to the Firebase backend using HTTPS.
    - In case of accidents or tampering or any other alerts are sent immediately to the web dashboard via Firebase for real-time monitoring.
    - The system caches data locally and will retry sending if there is no network signal, ensuring no data is lost.

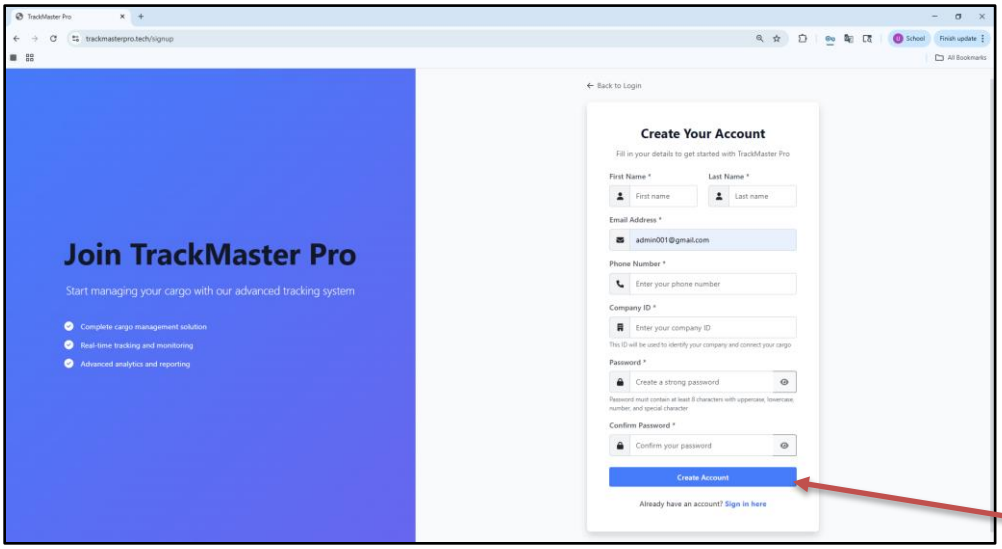
  - ### 3.4.3 Battery and Power Management

    - The device is powered directly from the vehicle's 5V supply.
    - A boost converter regulates the voltage up to 5V for stable operation of the ESP32 and sensors.
    - The included battery backup allows the device to keep running for a few hours even if the vehicle power is off, ensuring continuous monitoring during short power outages.

4. Administrator Web Application Guide

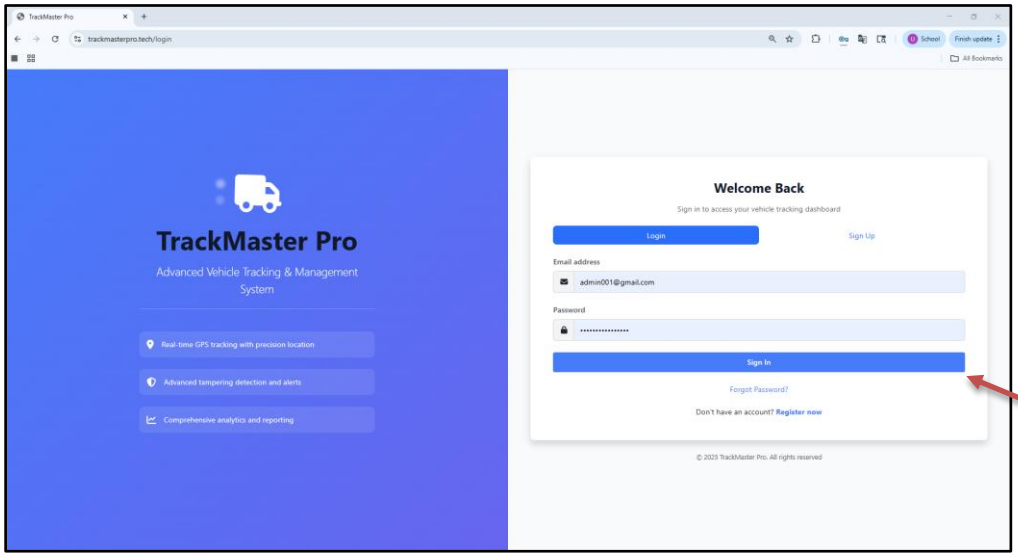
- 4.1 Getting Started
  - 4.1.1 First-Time Setup

    Admin Registration Page - Administrator needs to create a company account by filling in company details, admin email, and password.



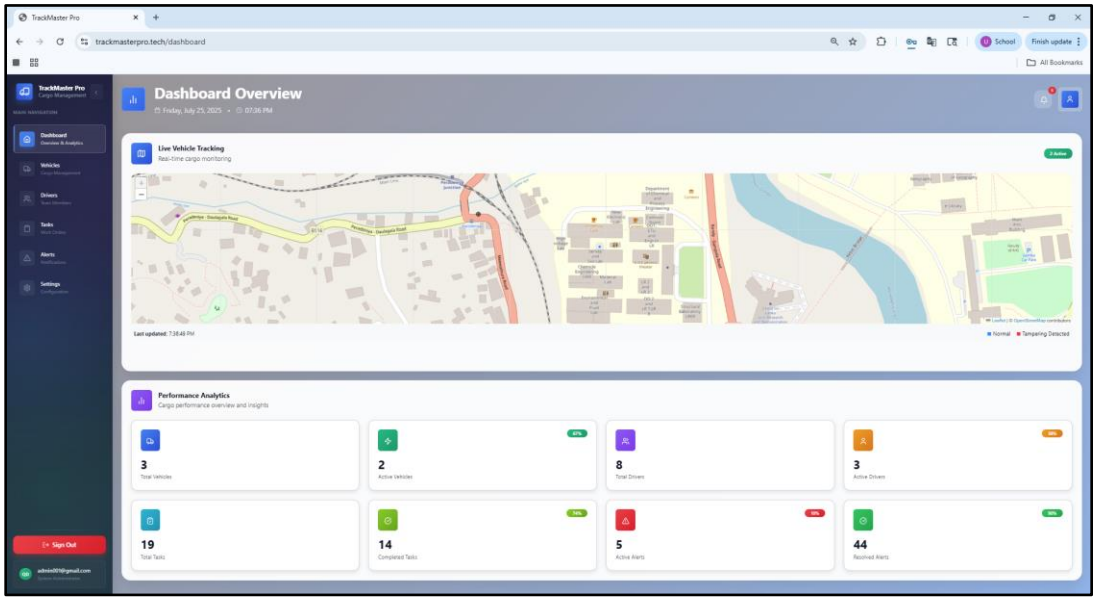
  - 4.1.2 Login Process

    Admin Login Page - After registration, administrator logs in using email and password.



4.1.3 Dashboard Overview

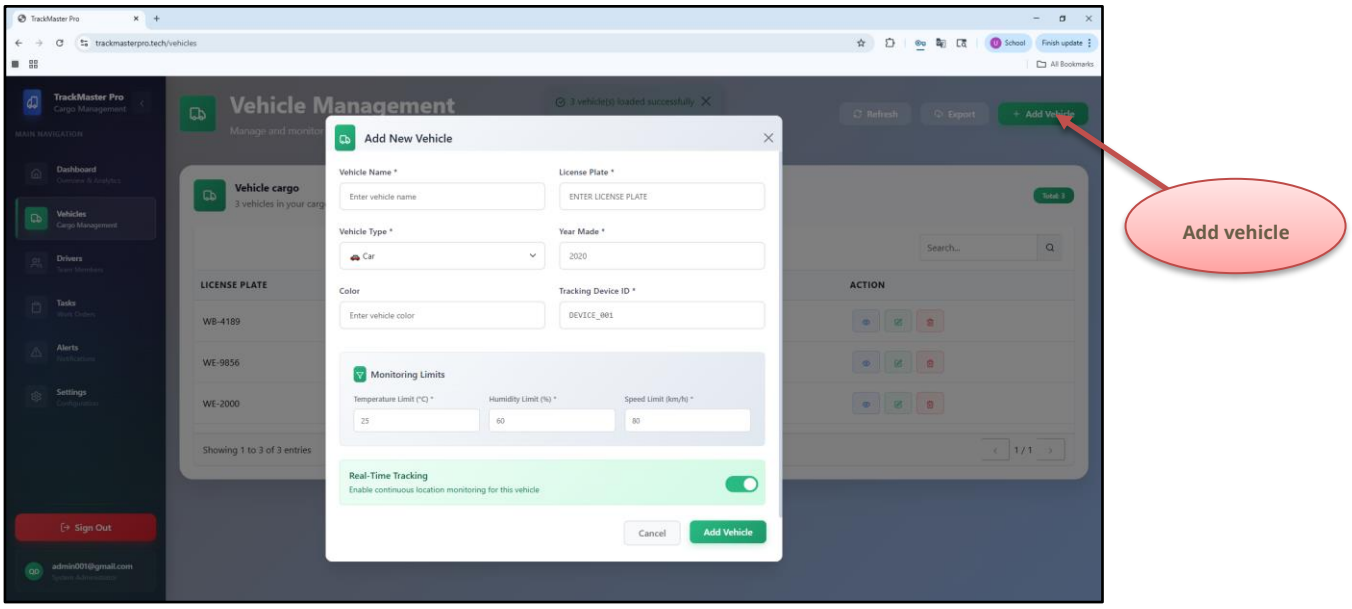
Admin Dashboard Home - Shows Cargo overview, active vehicles, alerts, and key performance metrics



4.2 Vehicle Management

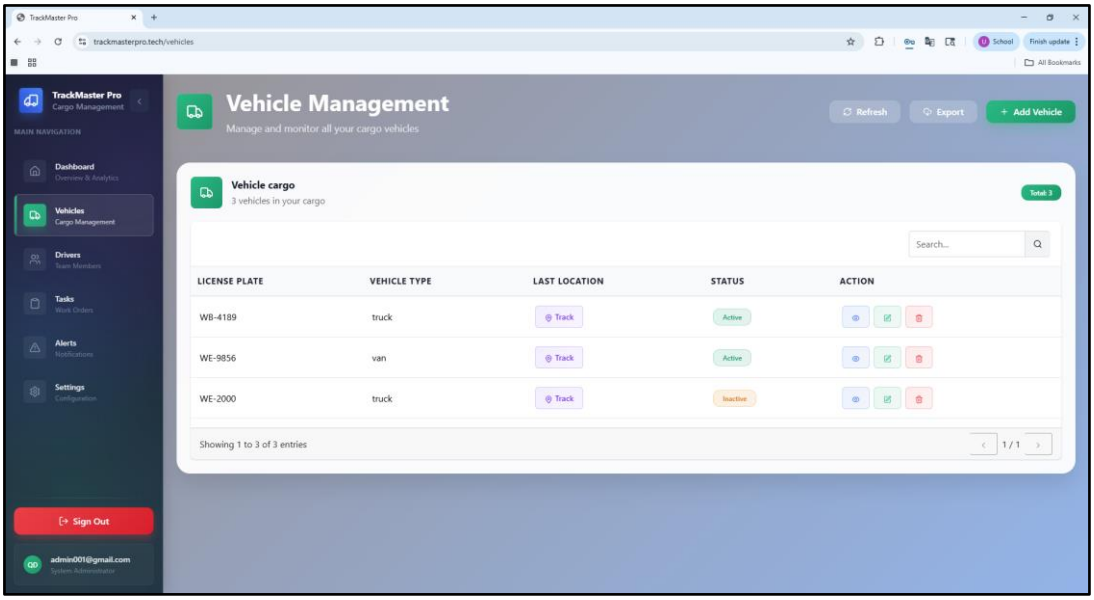
4.2.1 Adding New Vehicles

Add Vehicle Form - Administrator fills in vehicle details including license plate, model, and driver assignment and the IoT device id.



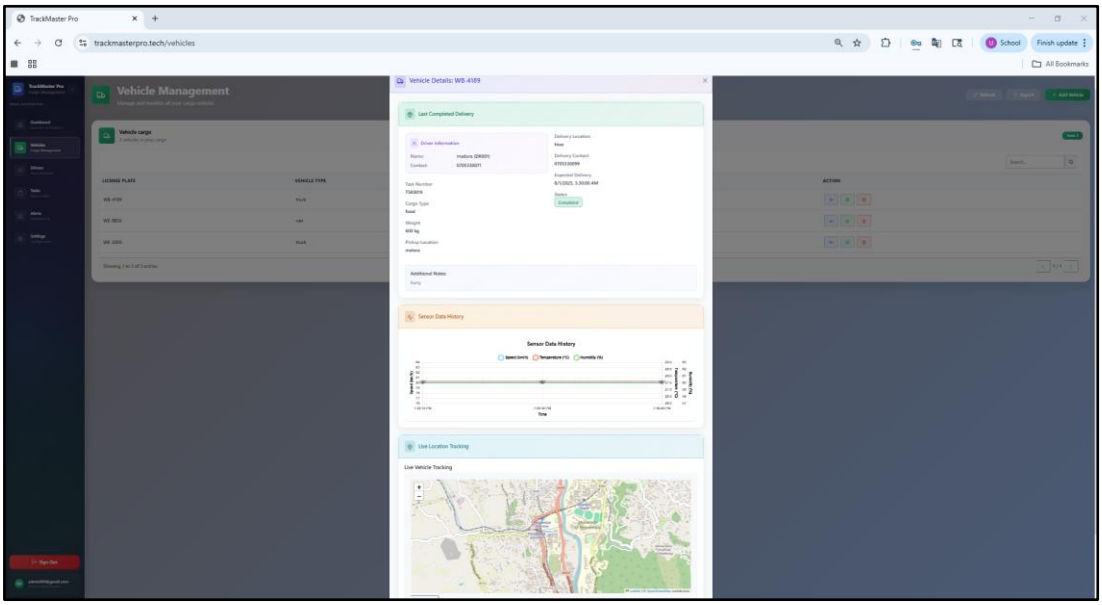
4.2.2 Vehicle Cargo Overview

Vehicle List Page - Displays all registered vehicles with status indicators (Active/Inactive)



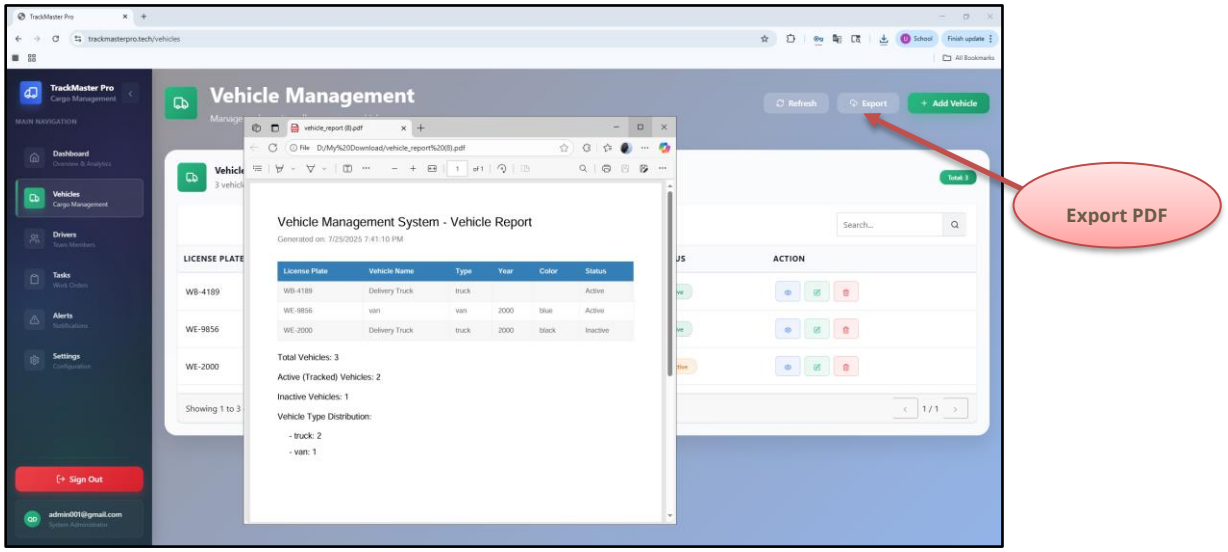
4.2.3 Real-Time Vehicle Tracking

Live Tracking Map – Click the “Track” button in the “Last Location” column to view real-time vehicle locations on an interactive map with status indicators.



4.2.4 Vehicle History and Reports

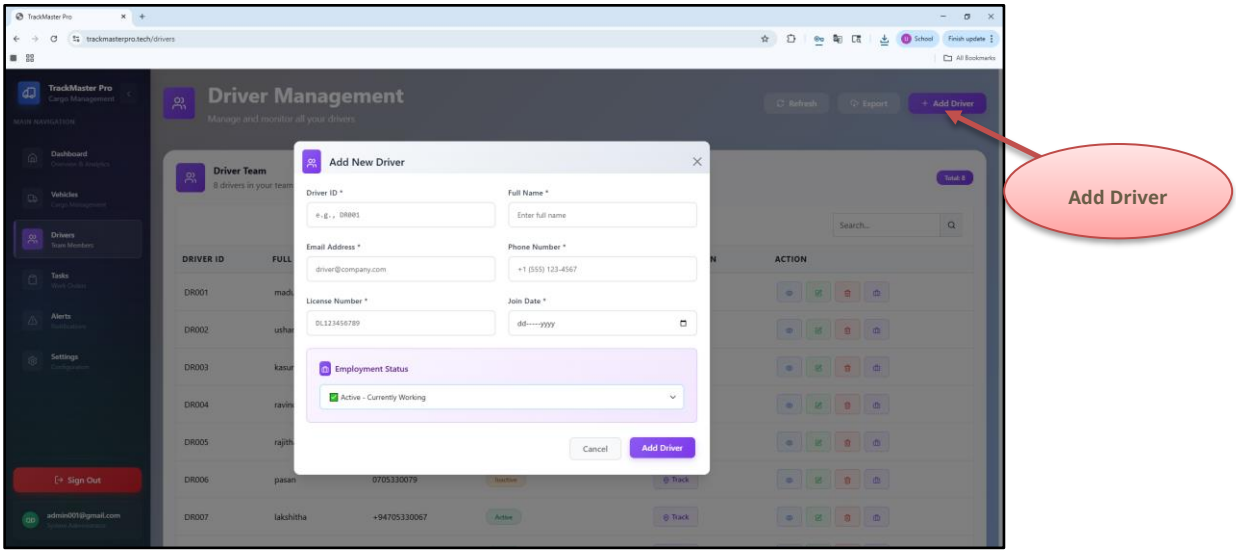
Vehicle History Page - Displays trip history, routes, and performance analytics for selected vehicle



4.3 Driver Management

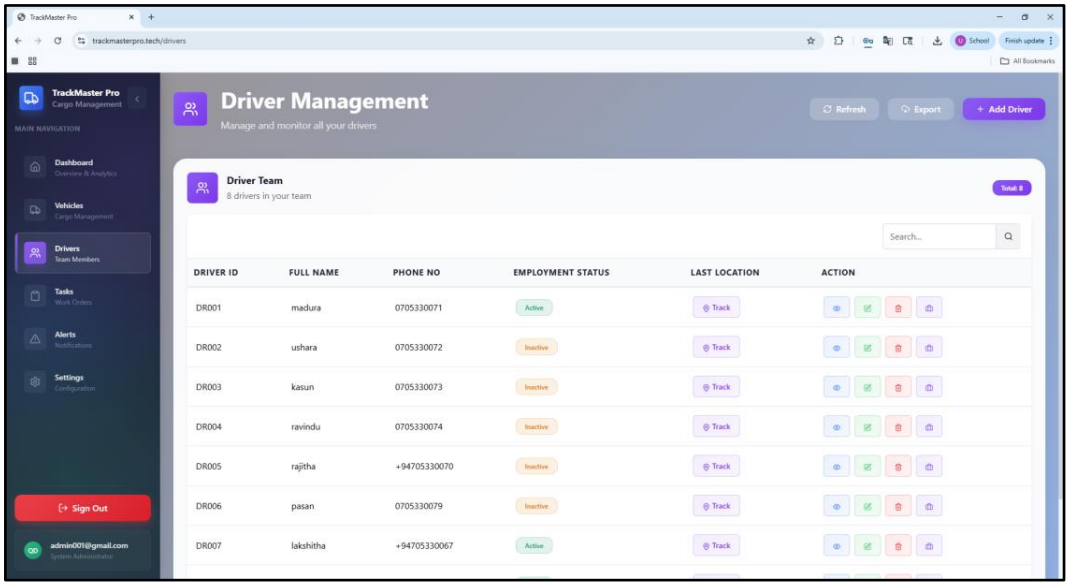
4.3.1 Driver Registration

Add Driver Form - Administrator creates driver accounts with personal details and credentials



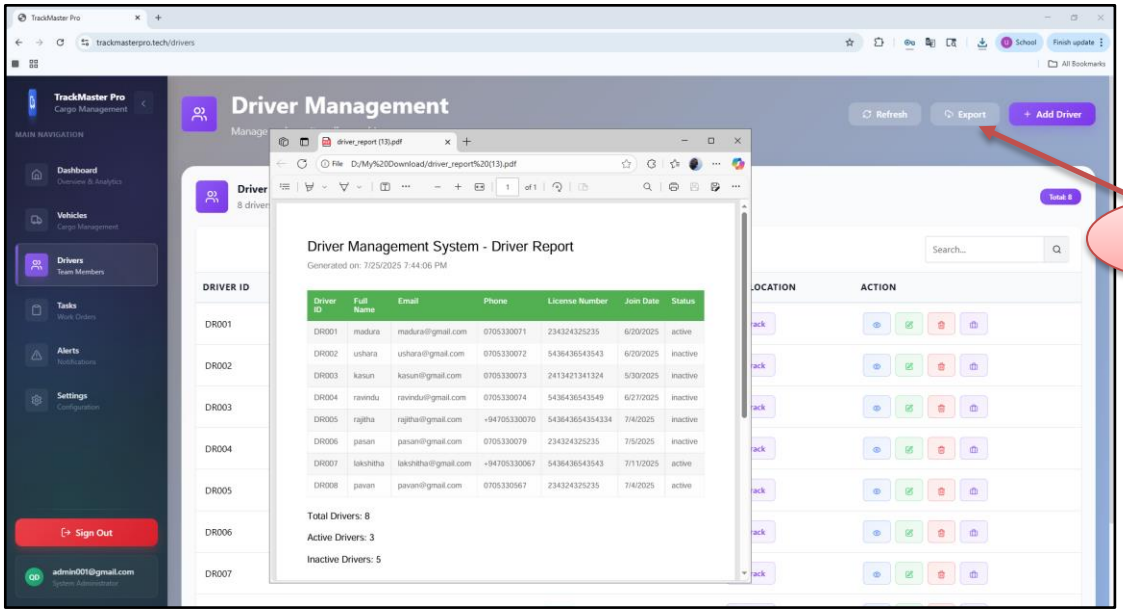
4.3.2 Driver Overview

Drivers List Page - Displays all registered Drivers with status indicators (Active/Inactive)



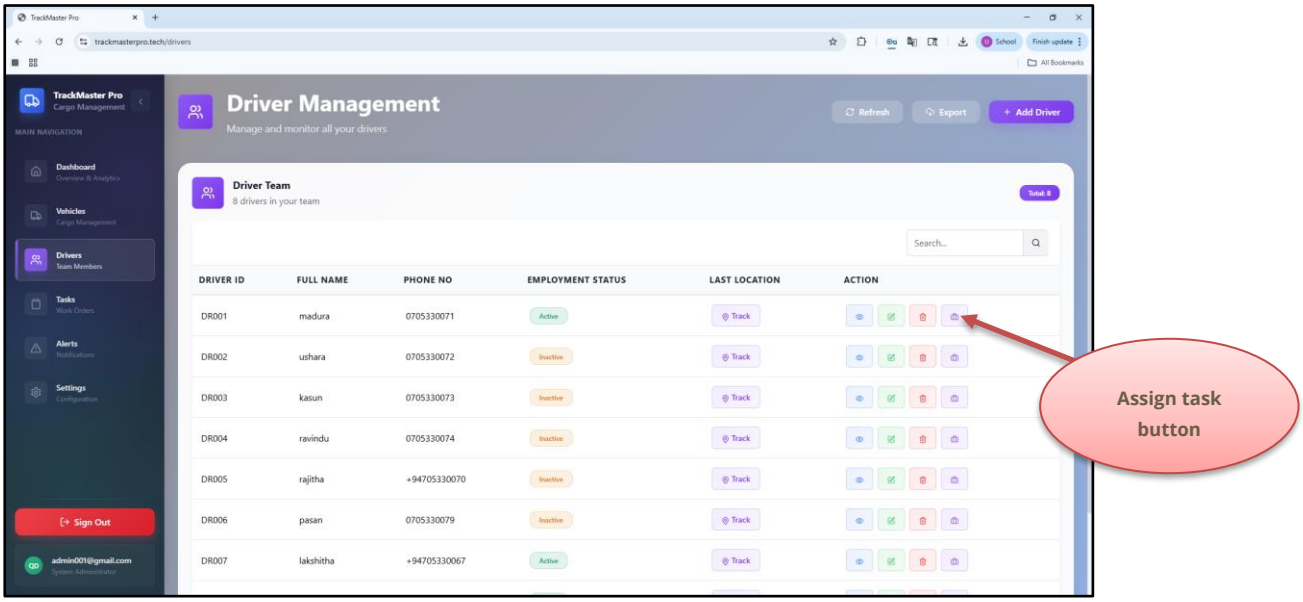
4.3.3 Driver Report

Driver Performance Dashboard - Shows total registered drivers, counts, and driver status.



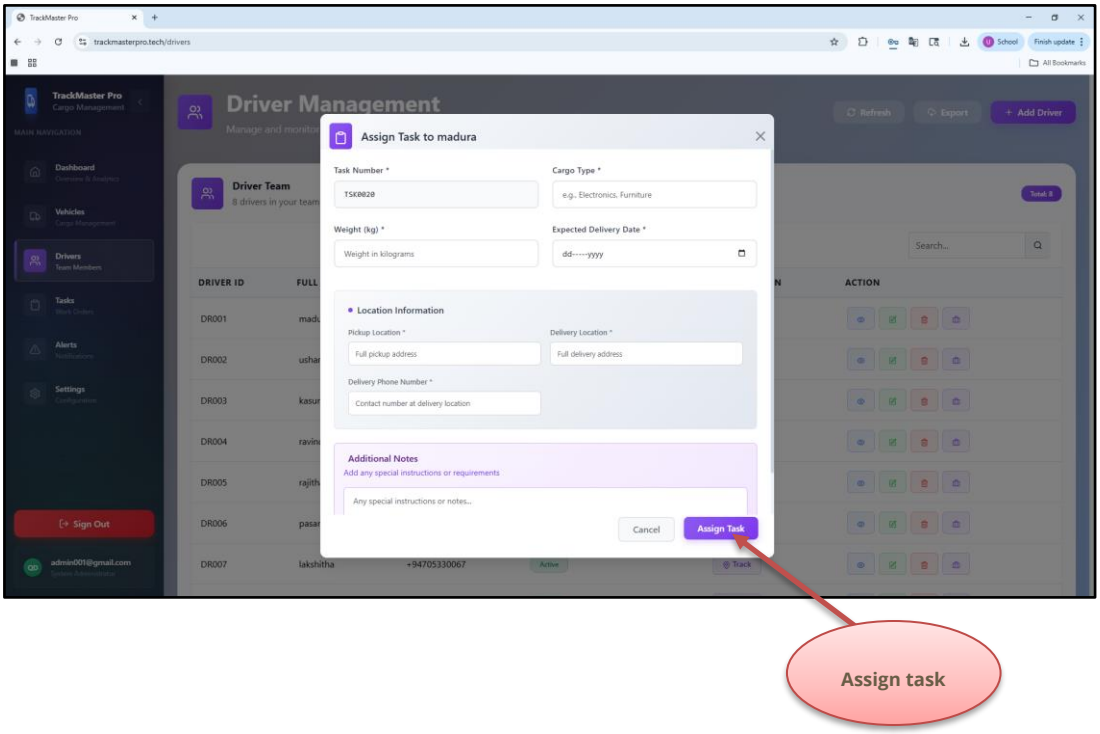
- 4.4 Task Management
  - 4.4.1 Task Assignment

    Task Assignment Interface - Assign tasks to specific drivers and vehicles



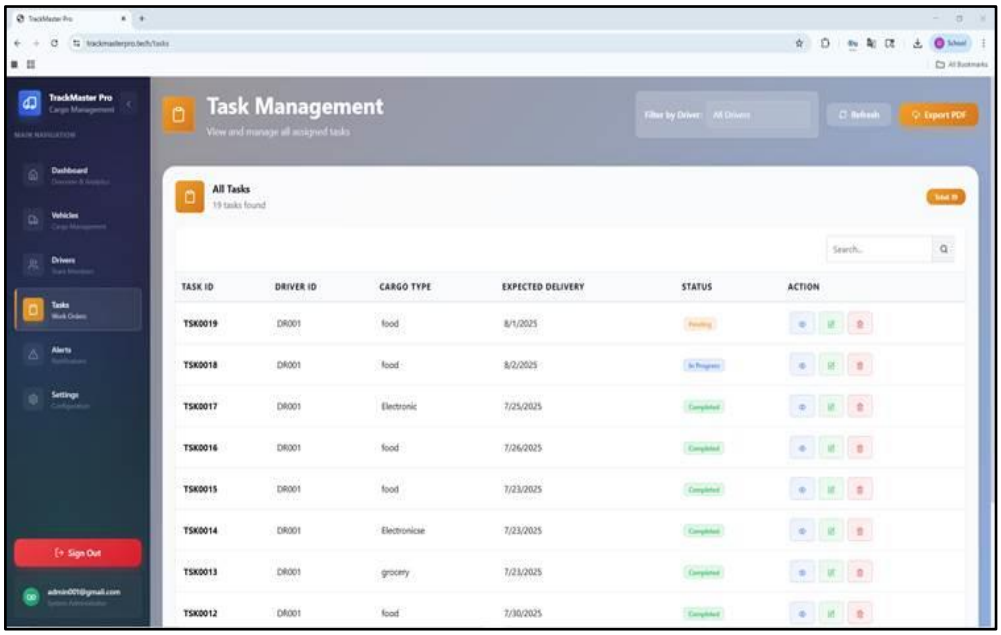
  - 4.4.2 Creating Tasks

    Add Task Form - Administrator creates tasks with destination, priority, and assignment details



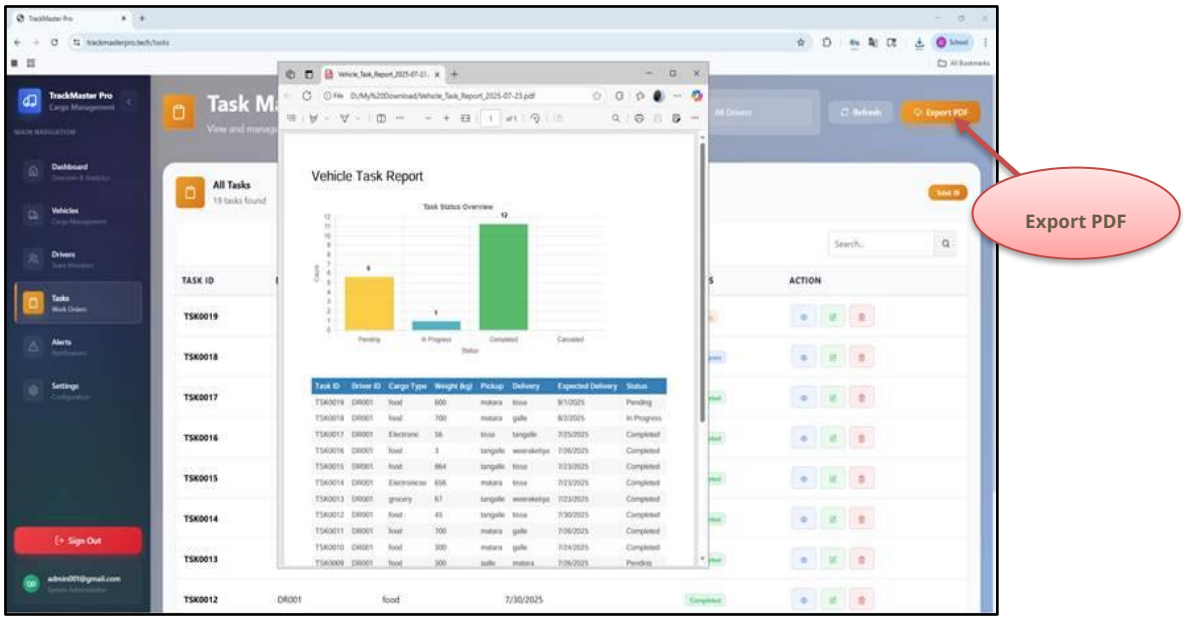
4.4.3 Task Monitoring

Task Status Dashboard - Monitor task progress and completion status



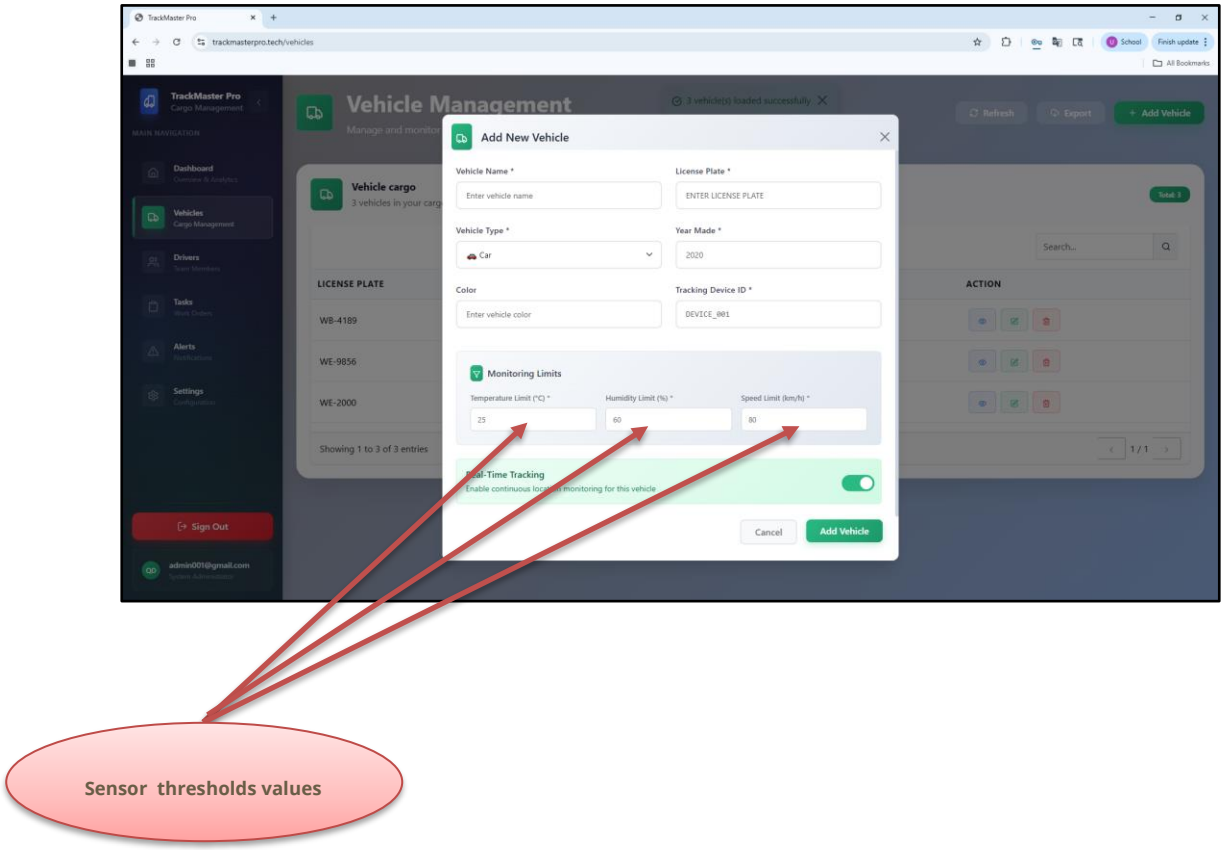
4.4.4 Tasks Report

task report



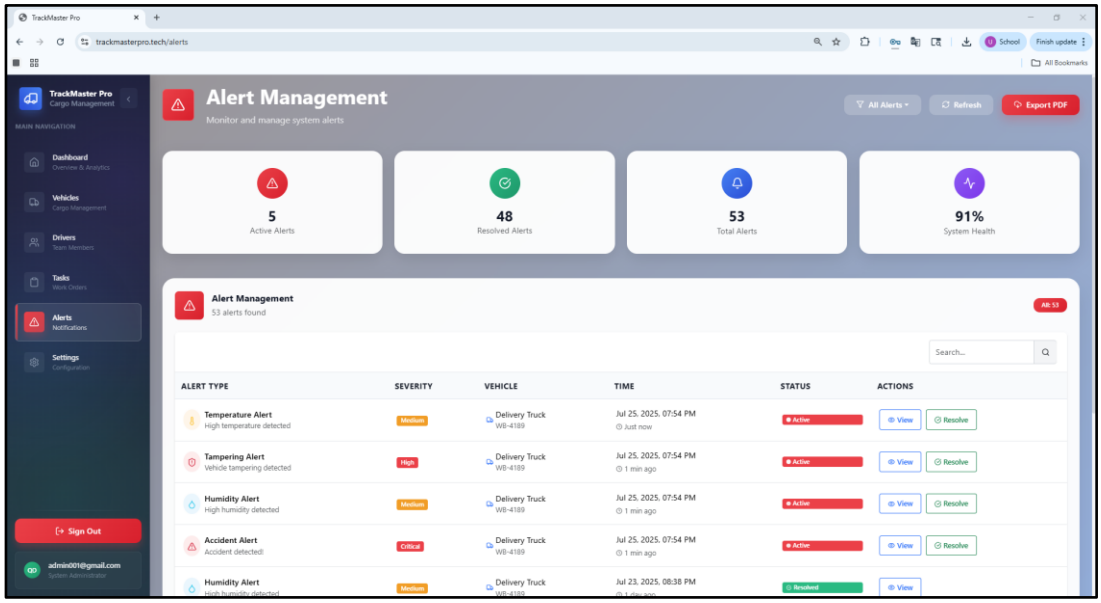
- 4.5 Alert and Notification Management
  - 4.5.1 Alert threshold adjust

    Edit specific vehicle number Details – Adjust thresholds for specific vehicles.



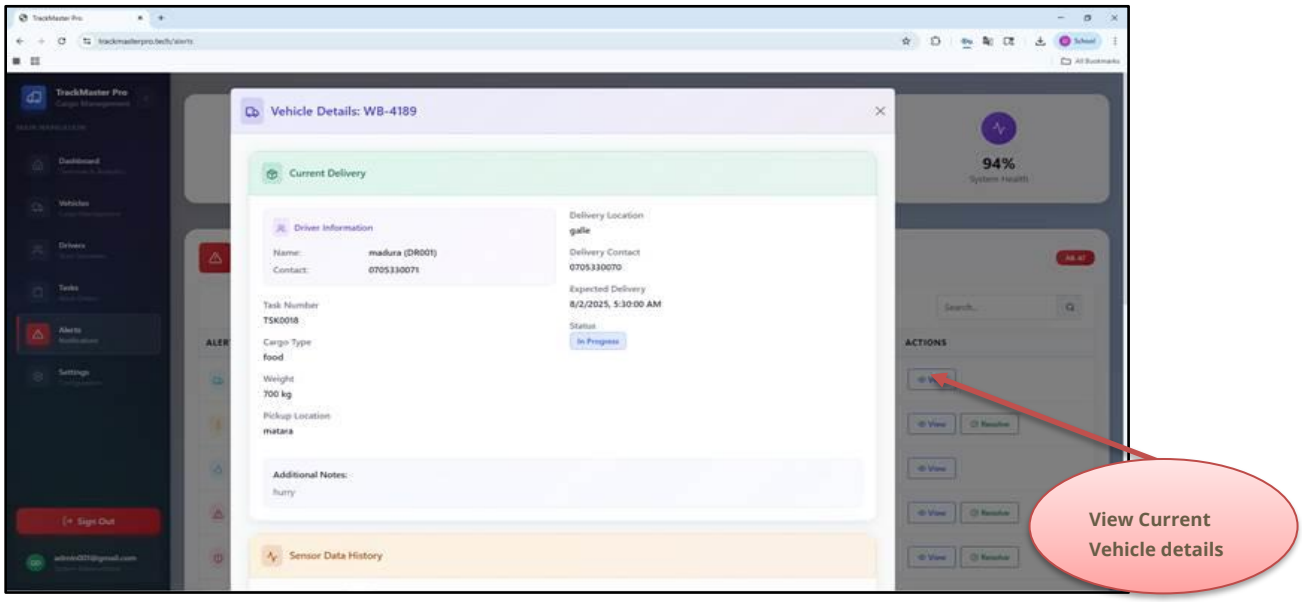
4.5.2 Real-Time Alerts

Alert Dashboard - Displays active alerts, including accidents, speeding, humidity, temperature, and unauthorized access (tampering).



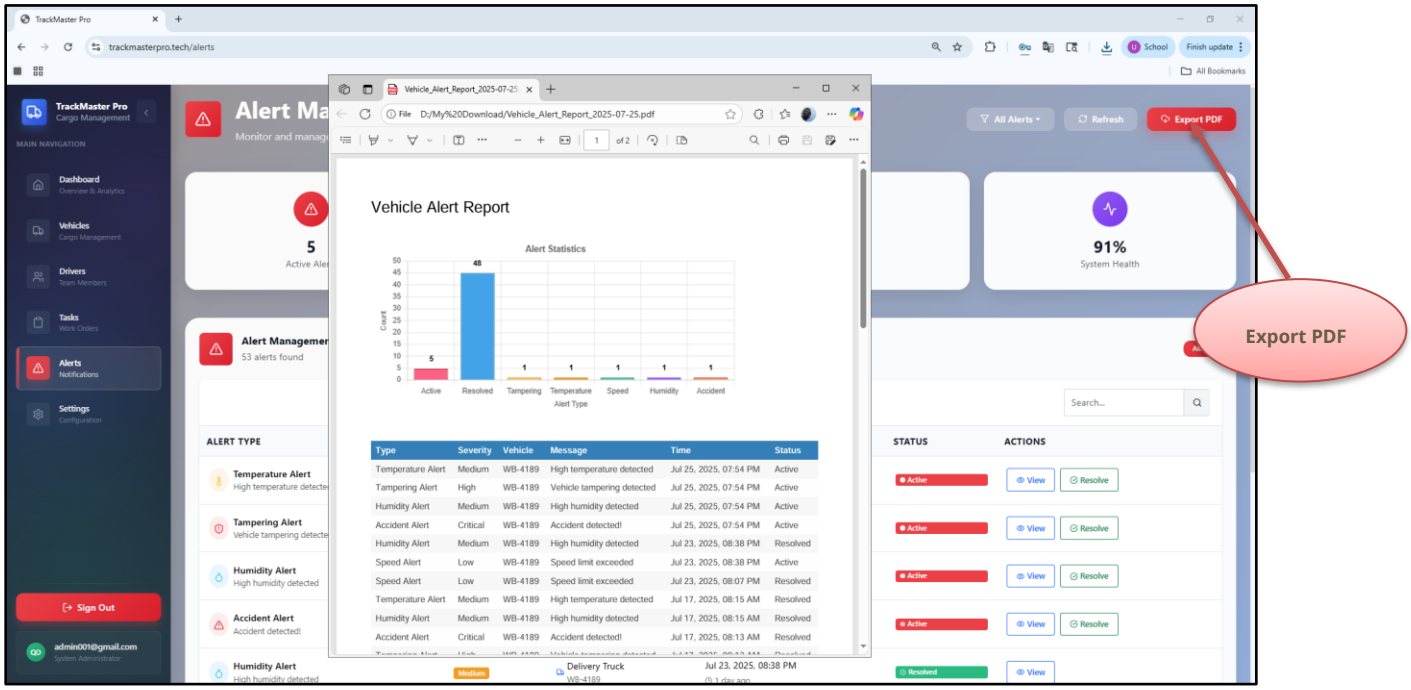
4.5.3 Emergency Response

Emergency Alert Interface - Displays emergency situations with driver contact and location details.



○ 4.5.4 Alerts Report genarate

Generates reports on alert history and resolution.

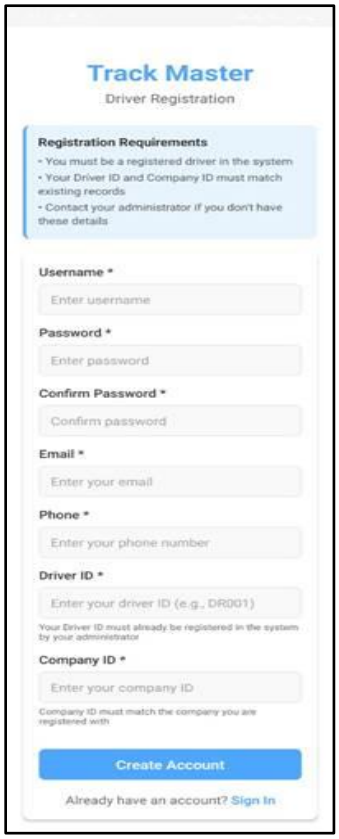


5. Driver Mobile Application Guide

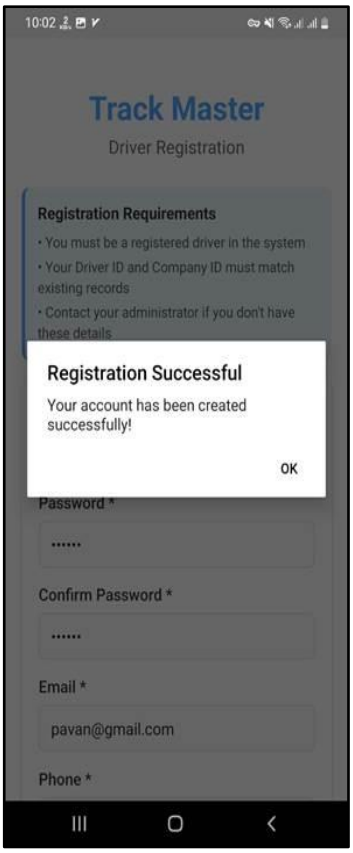
- 5.1 Getting Started
  - 5.1.1 App Installation

    Drivers download the app APK from the provided source.

  - 5.1.2 Driver Registration



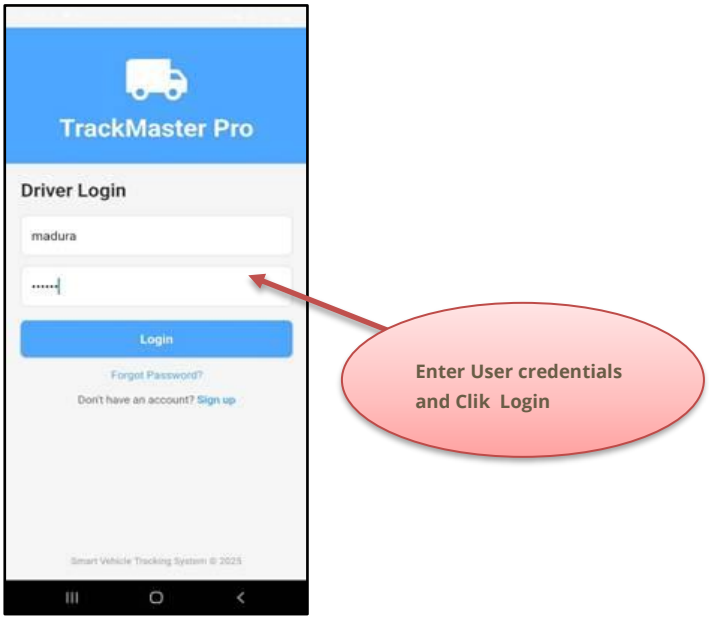




Mobile Registration Form - Drivers enter a registration code provided by the administrator and create a password. (Note: Driver details must be registered in the Driver Management section of the admin frontend before mobile app registration.)

5.1.3 Login Process

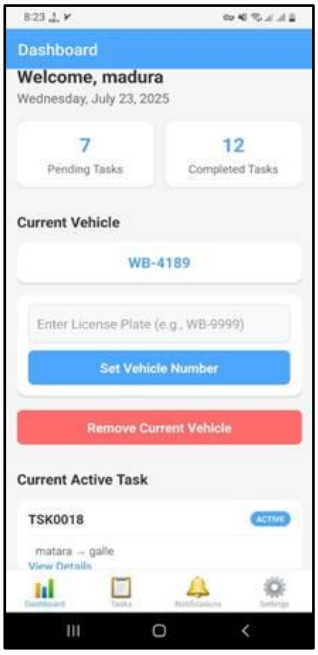
Mobile Login Screen - Drivers log in using their employee username and password.



5.2 Dashboard and Navigation

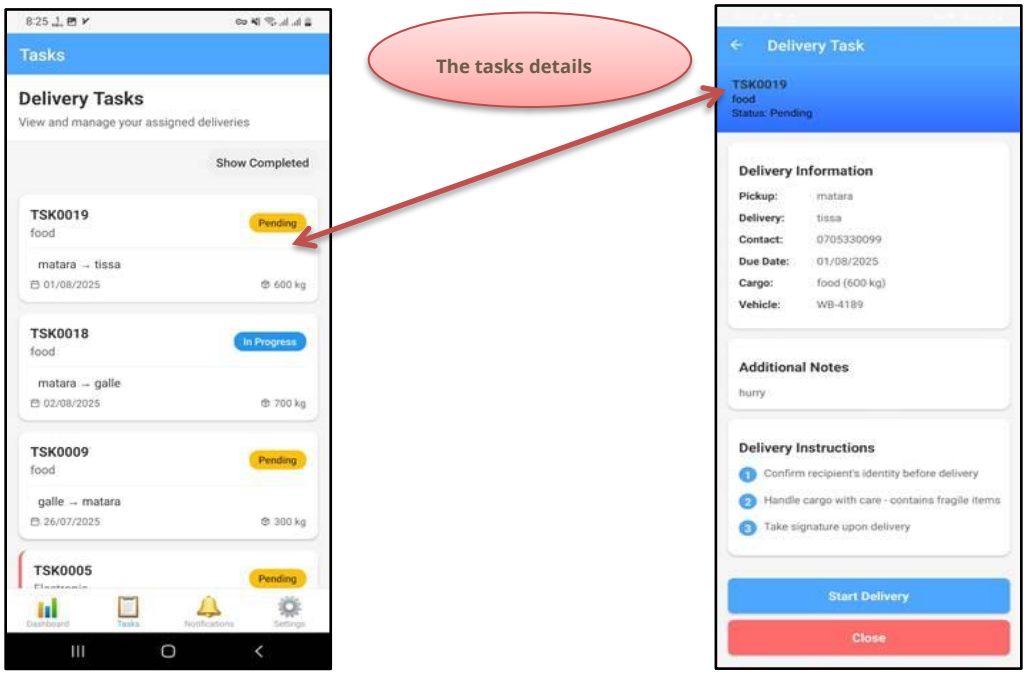
5.2.1 Home Dashboard

Mobile Dashboard - Shows assigned vehicle, current status, and daily tasks



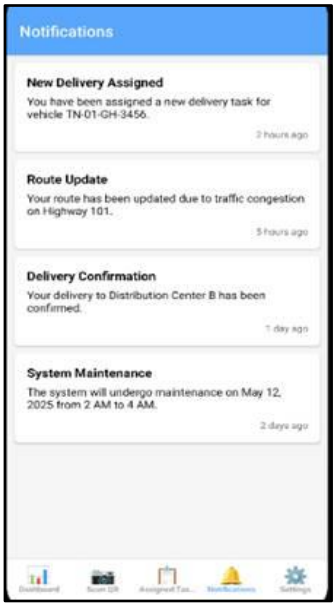
○ 5.2.2 Task List

Task List Interface - Shows tasks assigned by the admin. drivers can start tasks from this section.



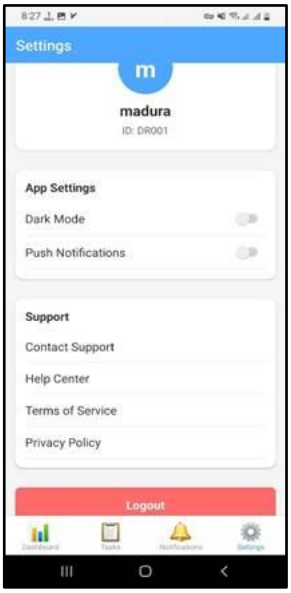
○ 5.2.3 Notifications

Notification Center - Displays new task assignments, task updates, and task deletions.



5.2.4 Profile Management

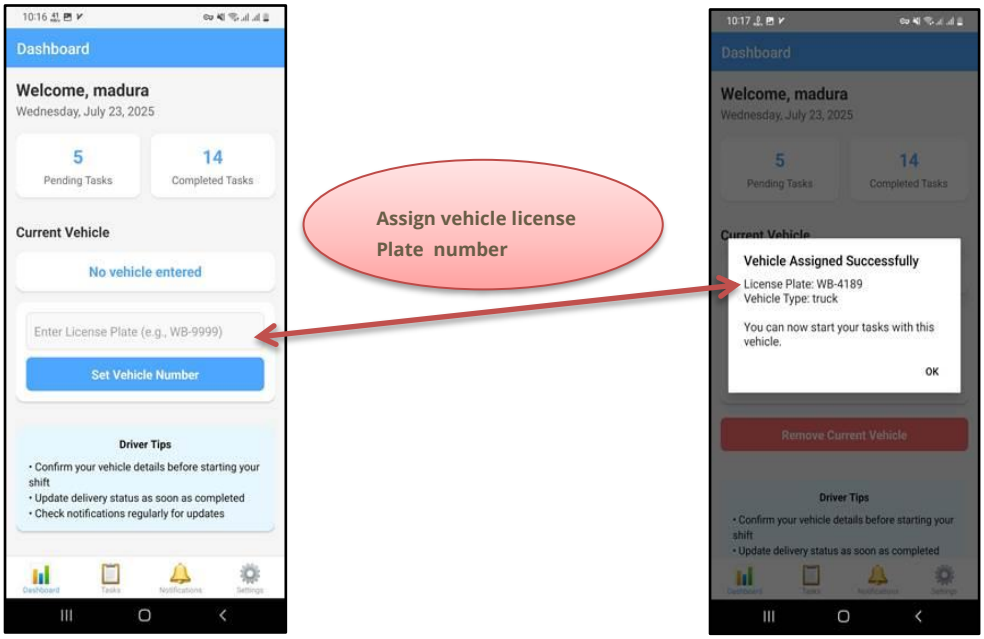
Driver Profile Screen - View and update personal information, change password



5.3 Vehicle Operations

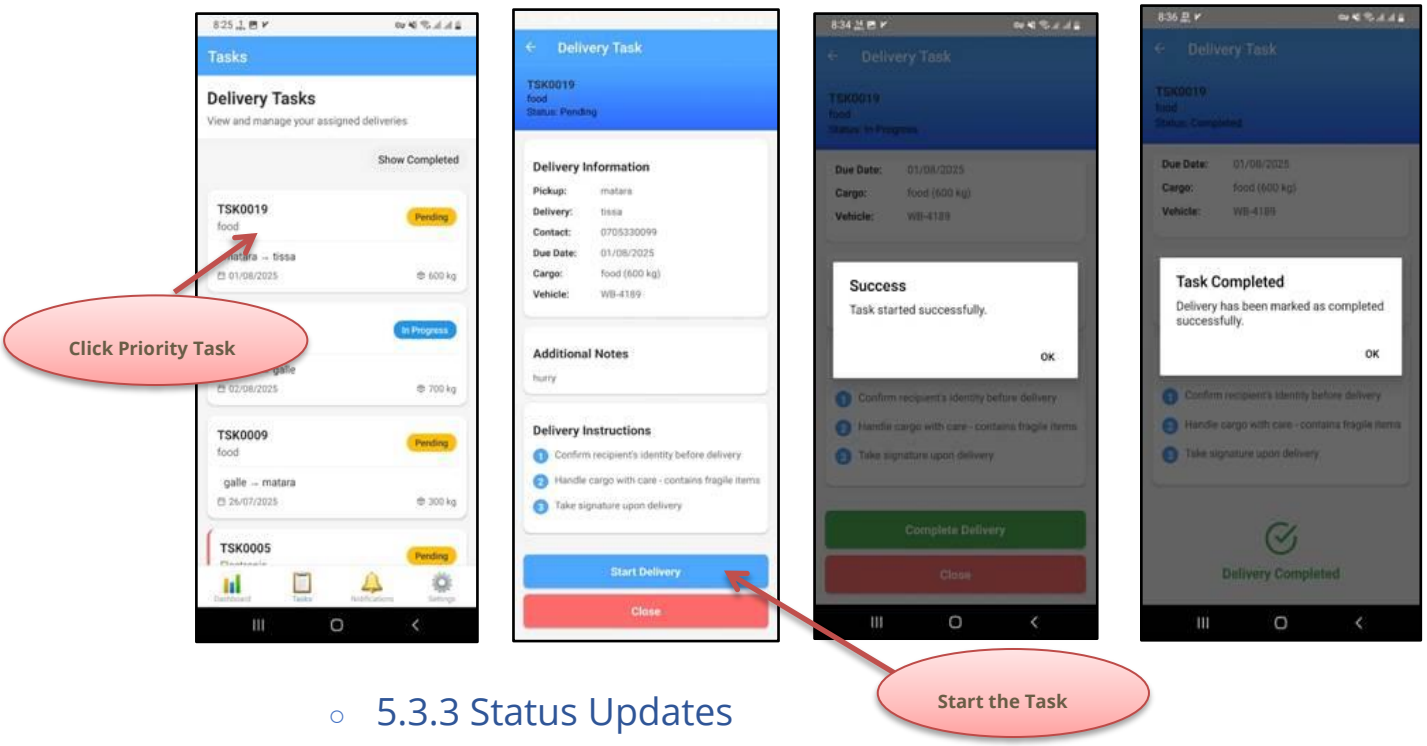
5.3.1 Vehicle Check-In/Check-Out

Vehicle Assignment Screen - Driver confirms vehicle assignment and starts/ends shift.



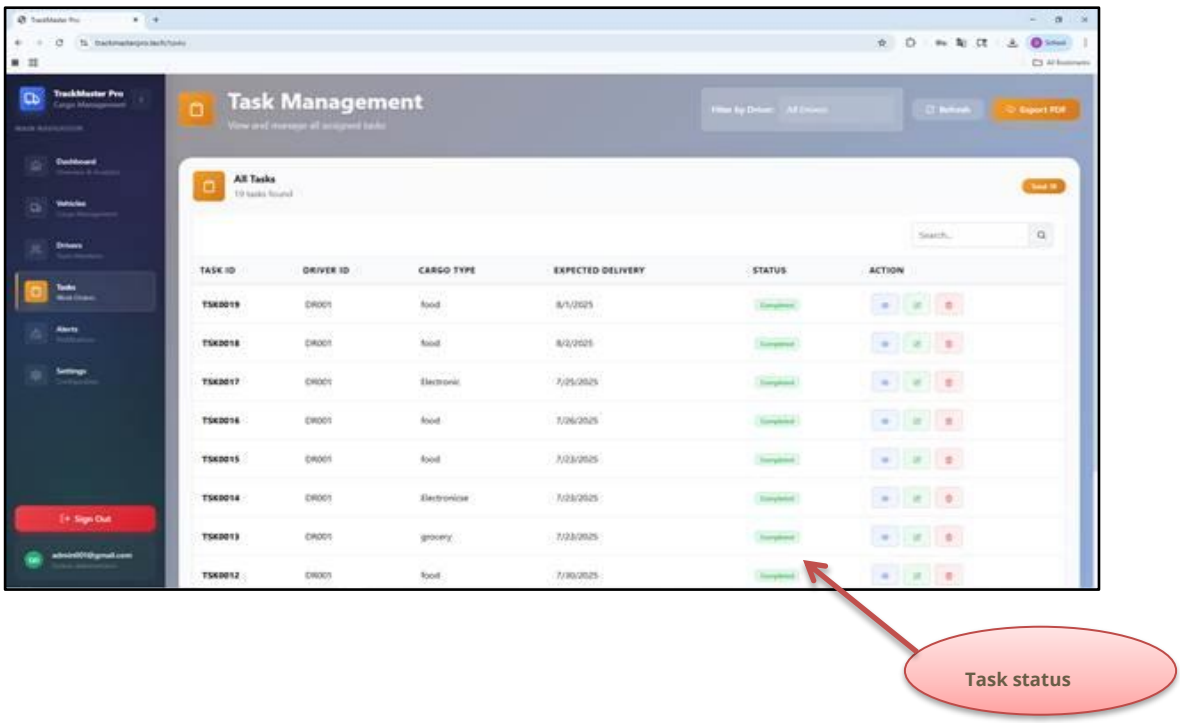
○ 5.3.2 Trip Management

- Trip Start Interface - Driver starts trip, enters destination, and confirms route



○ 5.3.3 Status Updates

Status Update Screen - Update trip status, report delays, or vehicle issues



- 5.4 Safety and Emergency Features
  - 5.4.1 Emergency Alert

    Emergency Contact - Contact admin or emergency services with one-touch emergency button

6. System Integration and Workflow

- 6.1 Complete System Operation

    - Admin Web App: Manages Cargo operations, creates tasks, monitors vehicles, and configures system settings
    - Driver Mobile App: Receives task assignments, updates trip status, and sends emergency alerts to administrators
    - IoT Device: Continuously collects GPS, sensor data, and transmits real-time information to cloud servers
    - Data Flow: Device → Cloud Database → Admin Dashboard → Task Assignment → Driver App → Status Updates

- 6.2 Data Synchronization

    - Real-Time Updates: Firebase Realtime Database synchronizes live GPS coordinates, speed, and sensor readings instantly
    - Cross-Platform Sync: Vehicle status, alerts, and task updates automatically reflect across web and mobile applications
    - Automatic Refresh: Dashboard displays live vehicle locations, driver status, and environmental data without manual refresh

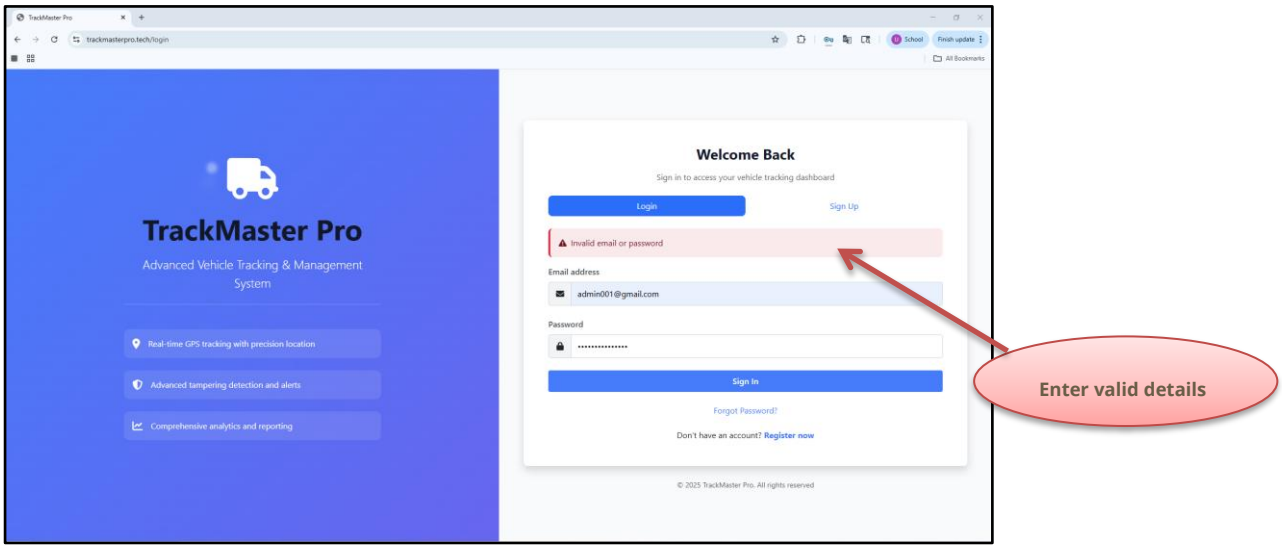
7. Troubleshooting Guide

- 7.1 IoT Device Troubleshooting

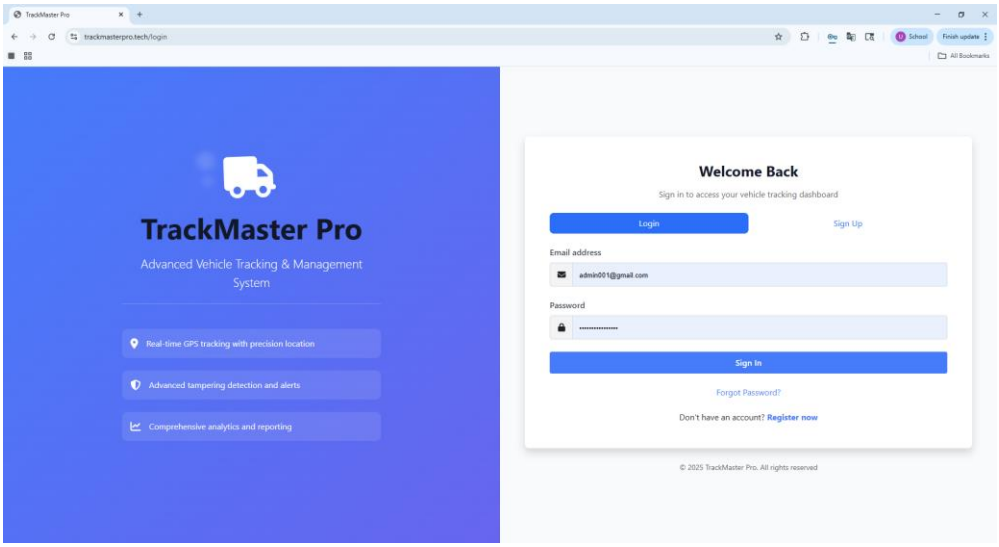
Issue	Possible Cause	Solution
Device Not Responding	Power supply issue or firmware crash	Check power connections, reboot device
GPS Signal Issues	Poor GPS antenna placement or obstruction	Reposition antenna for clear sky view
Communication Failures	SIM card not registered or poor GSM signal	Verify SIM activation,relocate device for better signal
Data Not Pushed to Firebase	Network issues, incorrect API keys, or server configuration errors	Check internet connectivity, verify Firebase credentials and API configuration; review error logs
Accident/Tamper Not Detected	Loose or faulty ADXL345 sensor connection; incorrect sensor calibration or thresholds	Check sensor wiring, recalibrate sensor thresholds; update firmware if needed

- 7.2 Administrator Web App Issues
  - 7.2.1 Login Problems

    Login Error Messages - Common login errors and solutions (e.g., invalid credentials). Verify credentials and try again, or clear the web cache and retry.



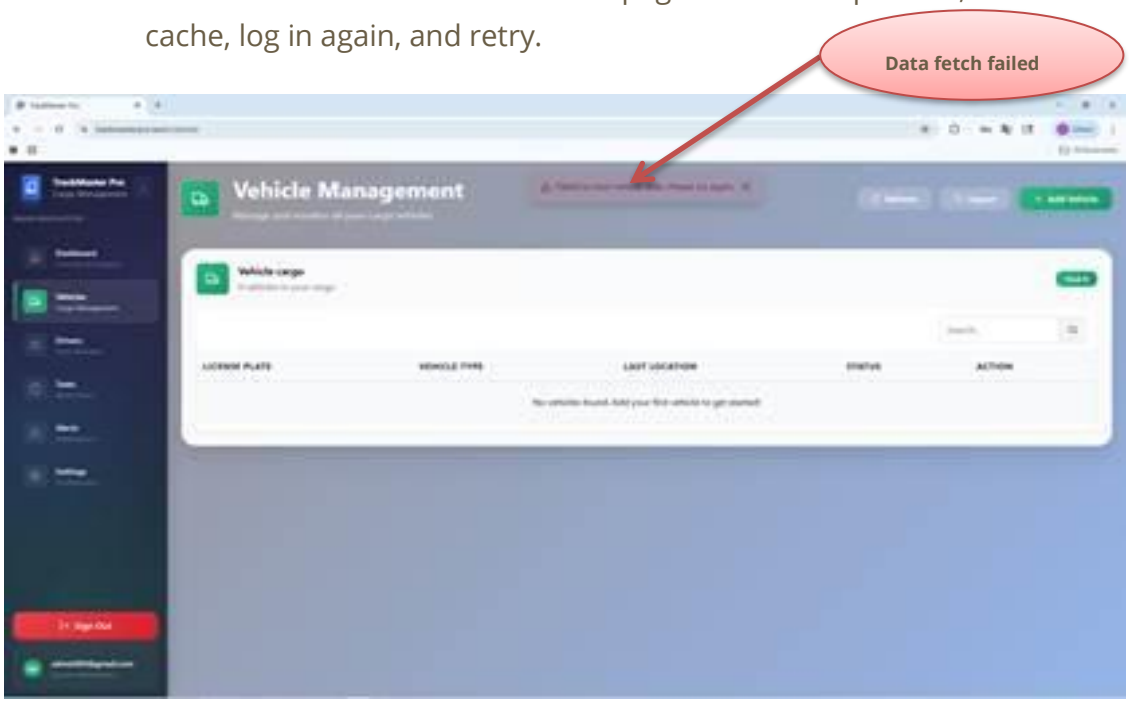
invalid credentials



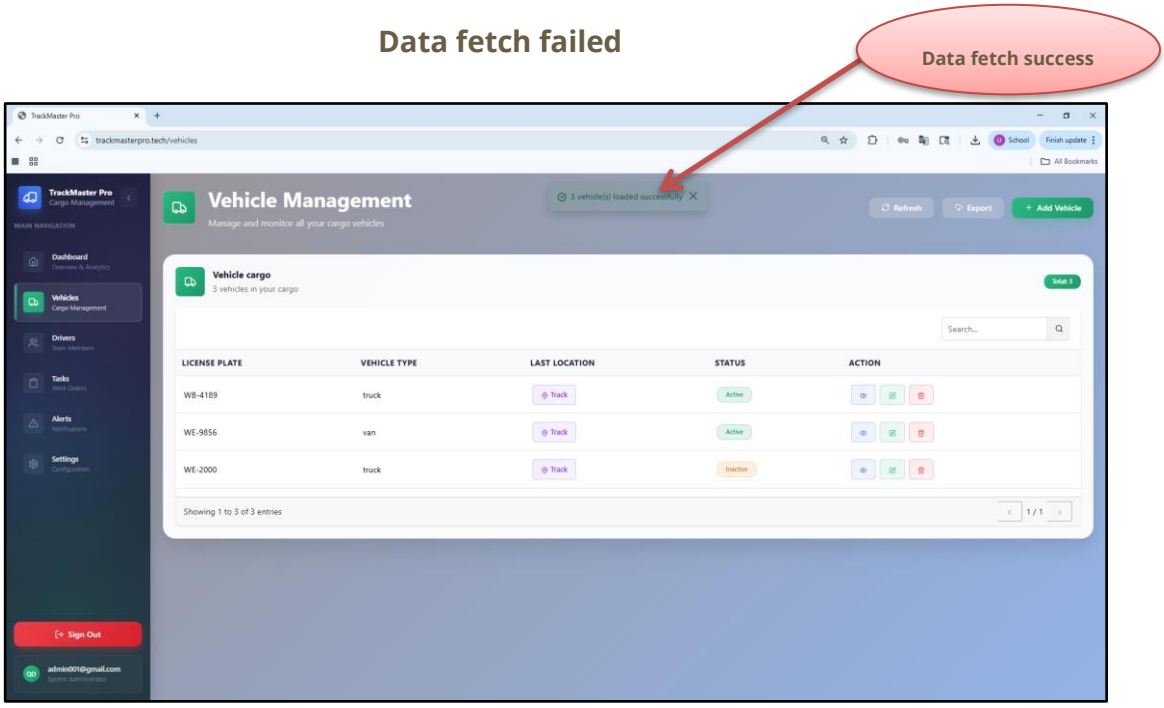
Valid credentials

7.2.2 Data Loading Issues

Loading Error Screen - When vehicle data or reports fail to load, check the internet connection and reload the page. If the issue persists, clear the web cache, log in again, and retry.



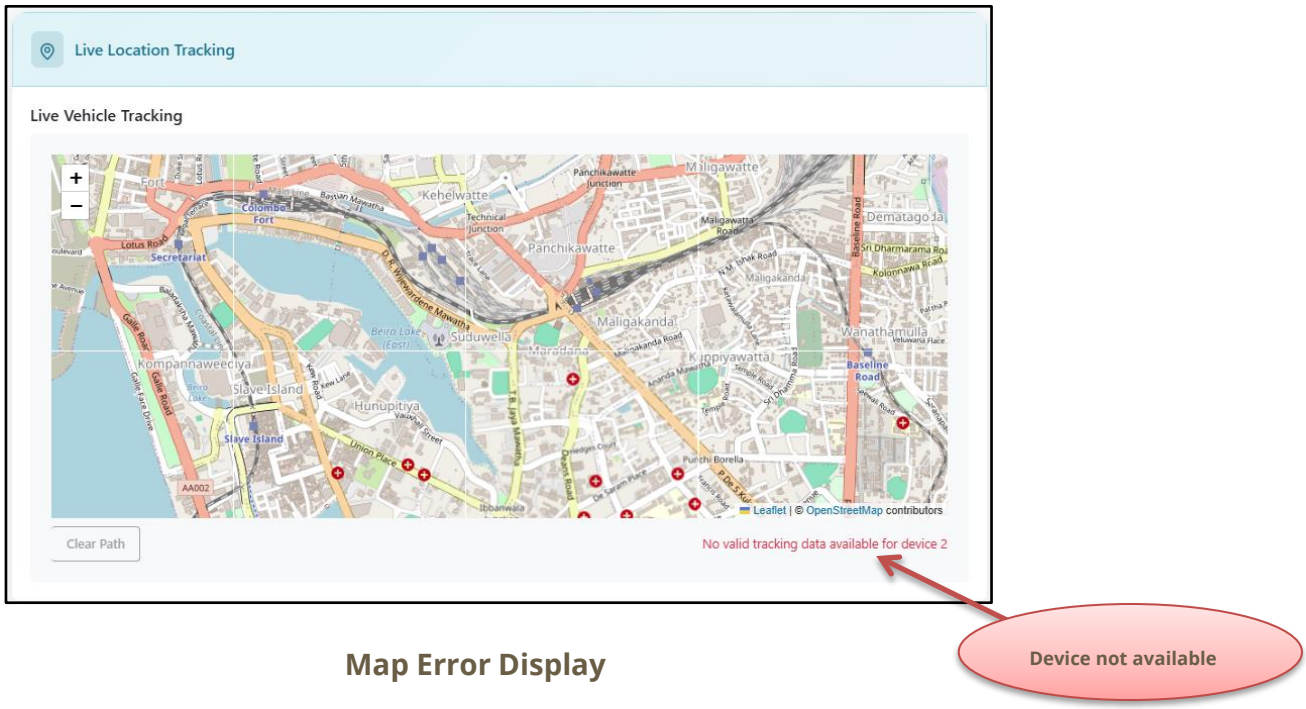
Data fetch failed



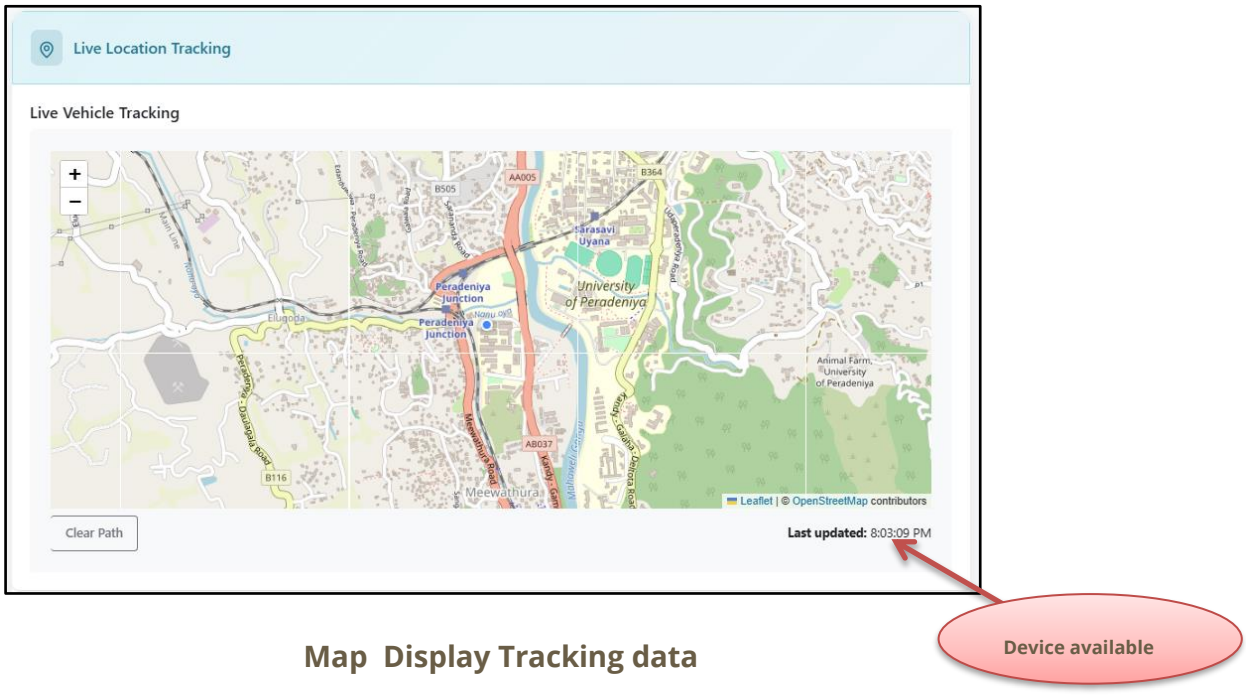
Data fetch Success

○ 7.2.3 Map and Tracking Issues

Map Error Display - For GPS tracking failures or map loading problems, ensure the device is powered on and connected to GPS. Check the internet connection, clear the web cache, and verify that the correct device ID is assigned to the vehicle during registration.



Map Error Display

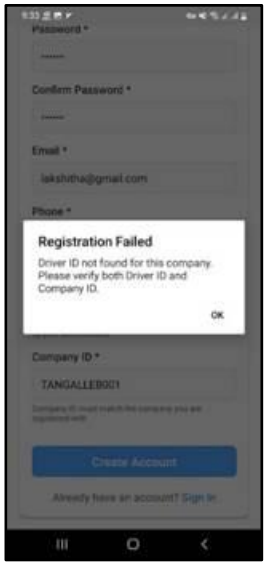


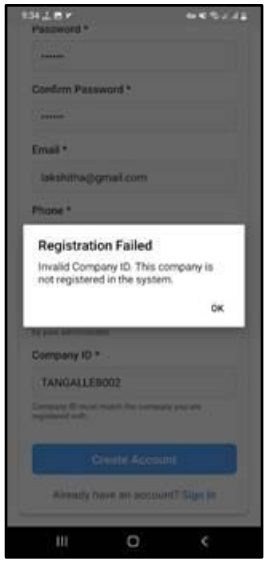
Map Display Tracking data

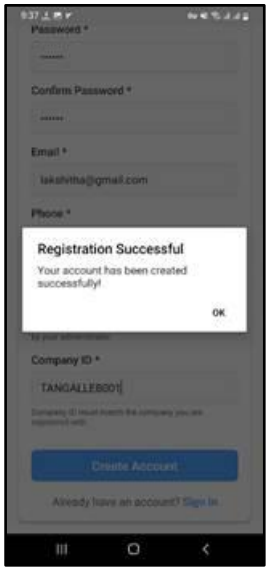
- 7.3 Mobile App Troubleshooting
  - 7.3.1 Registration Issues

    Registration Error Messages - For invalid codes or network errors during signup, verify the entered details and format. Ensure the driver's details are registered in the admin frontend before attempting mobile app registration. Contact the admin if issues persist.









Image(1) **image(2)** **image(3)** **Image(4)**

- **Image 1** – Error shown when the **Driver ID** is not entered in the correct format (e.g., DR007).
- **Image 2** – Registration failed because the driver was not registered first in the **Admin Registration** section.
- **Image 3** – Registration failed due to an **invalid or unregistered Company ID** entered in the **Company ID** field.
- **Image 4** – Registration completed **successfully**.

7.3.2 Connectivity Problems

Network Error Screen - For offline mode or connectivity issues, check for connection loss or weak signal and attempt to log in using a stable connection.

7.3.3 Data load

if auto refresh is fail try manual reload.

8. Maintenance and Updates

- ## 8.1 Regular Maintenance Tasks.

  Ensure the device is powered on before starting daily tasks. If not, contact the admin to report the issue.

- ## 8.2 Software Updates

  If the server is down, contact the development team and share details.

  Check for the latest version of the mobile app, as new versions include new features and bug fixes.

- ## 8.3 Hardware Maintenance

  Regularly inspect the IoT device for physical damage and ensure secure mounting.

9. Security and Privacy

- 9.1 Data Protection
    - All data transmitted using HTTPS/TLS 1.3 encryption
    - Passwords hashed with dcrypt, never stored in plain text
    - Role-based access control with strict user permissions
- 9.2 Password Security
    - Minimum 8 characters with mixed case, numbers, and special characters
    - Use strong passwords for both admin and driver accounts to enhance security
    - Account lockout after 5 failed login attempts
- 9.3 Privacy
    - Do not share cargo information or other sensitive details with third parties
    - Driver personal information accessible only to authorized personnel
    - System complies with data protection regulations (GDPR, local laws)

10. Frequently Asked Questions (FAQ)

- 10.1 General Questions

  - **Why do we use “TrackMaster Pro”?**
    TrackMaster Pro provides a comprehensive solution for real-time vehicle tracking, cargo monitoring, and operational efficiency, tailored for small and medium-sized businesses.
  - **Are the cargo details secure?**
    Yes, cargo details are protected with encryption and strict access controls to ensure confidentiality.
  - **How does the system improve operational efficiency?**
    The system optimizes cargo management through real-time tracking, task assignment, and detailed analytics, reducing downtime and costs.
  - **What happens if the system experiences downtime?**
    Contact the technical support team immediately to report and resolve the issue.

10.2 Technical Issues

- **What should I do if the device is not powered on or not connected?**
  Check the power supply and connections. Ensure the device is securely installed and contact the admin if the issue persists.
- **What should I do if the mobile app or admin app is not working correctly?**
  Verify the internet connection, clear the app or browser cache, and ensure the latest app version is installed. Contact support if the issue continues.
- **Why is the GPS signal weak or unavailable?**
  Check the device's antenna positioning and ensure it is not obstructed. Relocate the device if necessary or contact support.
- **What should I do if alerts are not being received?**
  Verify notification settings in the app and ensure a stable internet or GSM connection. Contact the admin to check alert configurations.

10.3 Account and Access Management

- **How do I reset my password?**
  Use the "Forgot Password" option on the login screen to receive a reset link via email.
- **What should I do if my account is locked?**
  Contact the admin or technical support to unlock the account after verifying your identity.

11. Support and Contact Information

- 11.1 Technical Support Contacts

TrackMaster Pro Development Team:

Email: e20456@eng.pdn.ac.lk
Email: e20272@eng.pdn.ac.lk

- 11.2 Emergency Support.

Sri Lanka Police: 119 (for accidents or tampering)
Suwasariya Ambulance Service: 1990 (for accidents)